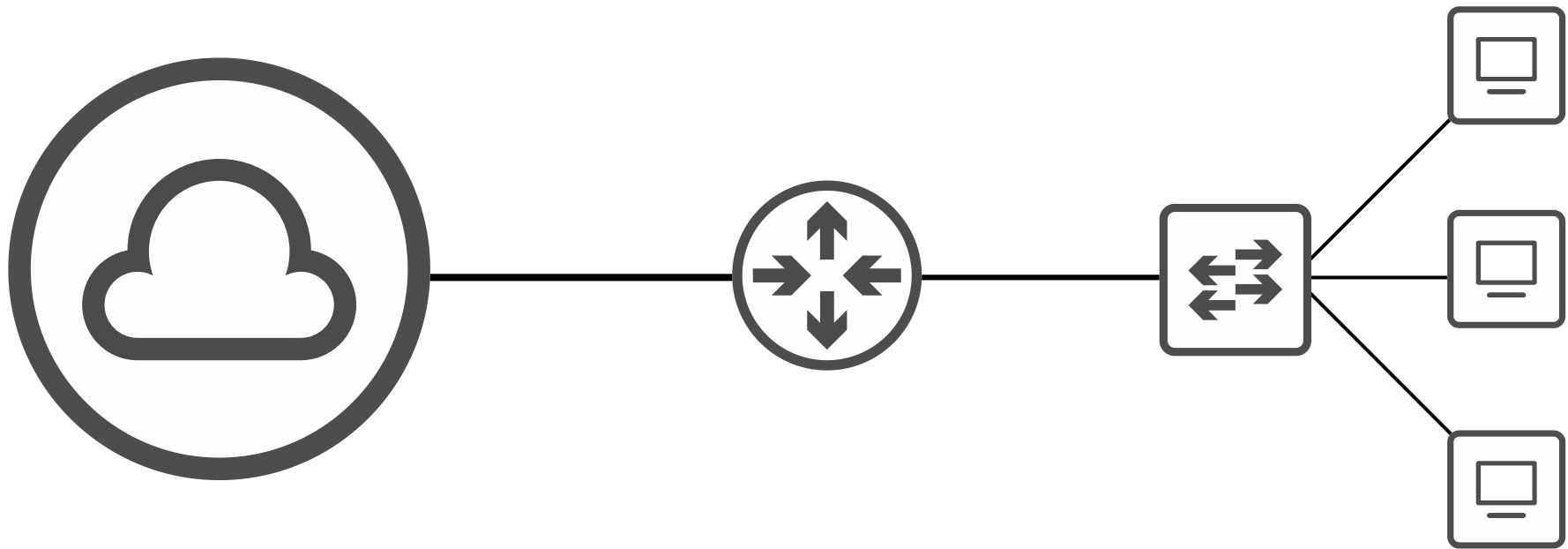


CCNA

Virtualization: Containers

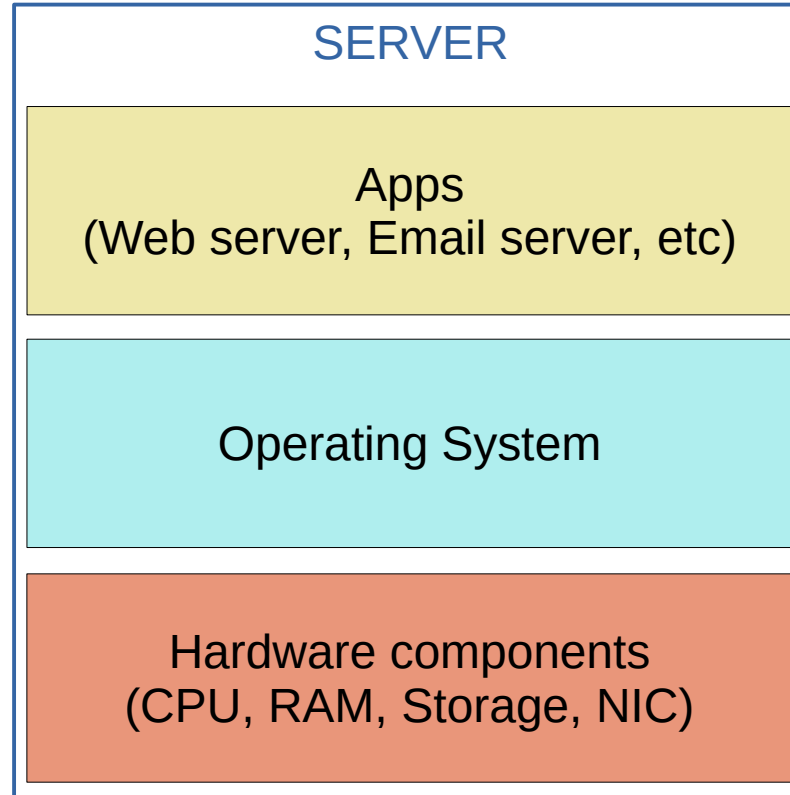


Things we'll cover

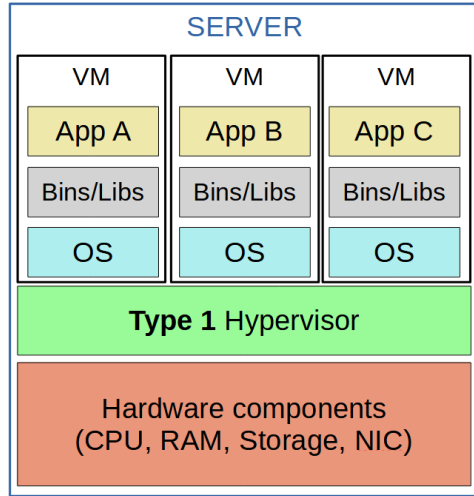
- Review of Virtual Machines
 - Type 1/Type 2 Hypervisors
- Containers (ie. Docker)
- Virtual Machines vs. Containers

Virtual Machines

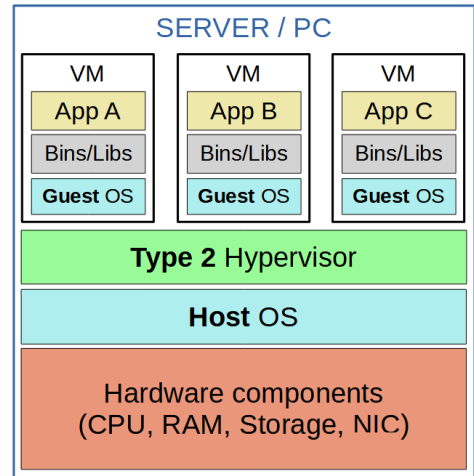
Apps running on a server without virtualization:



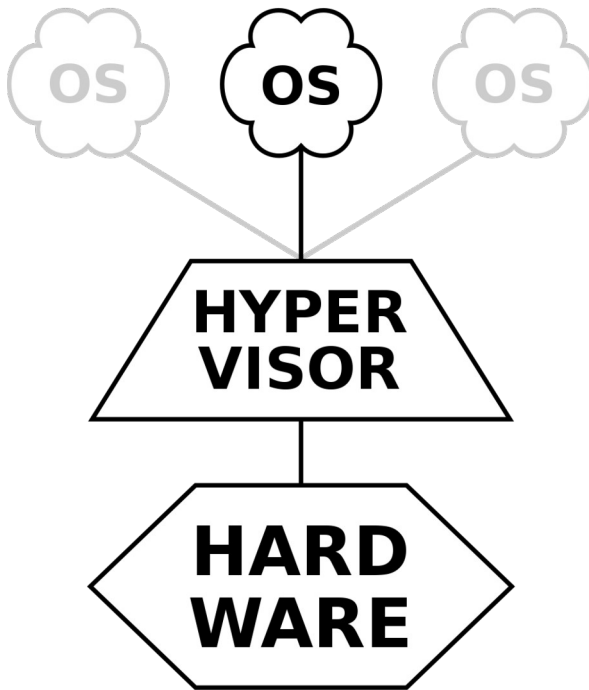
Virtual Machines



- **Virtual Machines** (VMs) allow multiple OS's to run on a single physical server.
- A **Hypervisor** is used to manage and allocate hardware resources to each VM.
 - **Type 1** Hypervisors (aka *Native* or *Bare-metal*) run directly on top of hardware.
 - **Type 2** Hypervisors (aka *Hosted*) run on top of a *Host OS* (ie. Windows).
- **Type 1** Hypervisors are widely used in data center environments.
- **Type 2** Hypervisors are commonly used on personal devices.
 - ie. running a virtual network lab on your PC using Cisco Modeling Labs (CML).
- The OS in each VM can be the same or different (Windows, Linux, macOS, etc).
- *Bins/Libs* are the software libraries/services needed by the Apps running in each VM.
- A VM allows its app/apps to run in an isolated environment, separate from the apps in other VMs.
- VMs are easy to create, delete, move, etc.
 - A VM can be easily saved and moved between different physical servers.

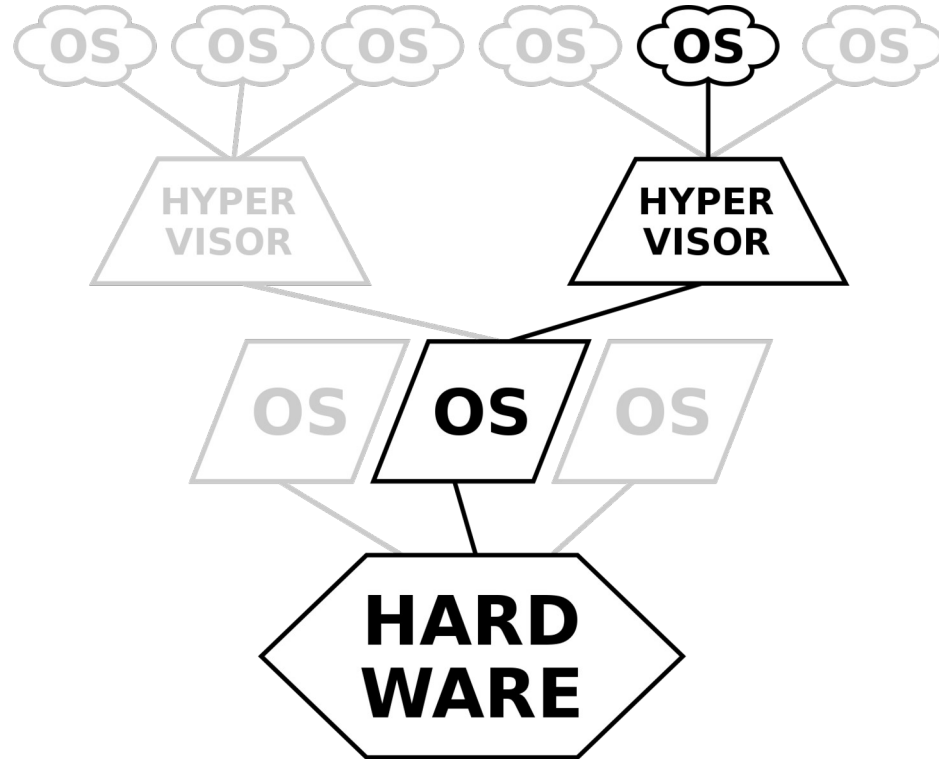


Virtual Machines



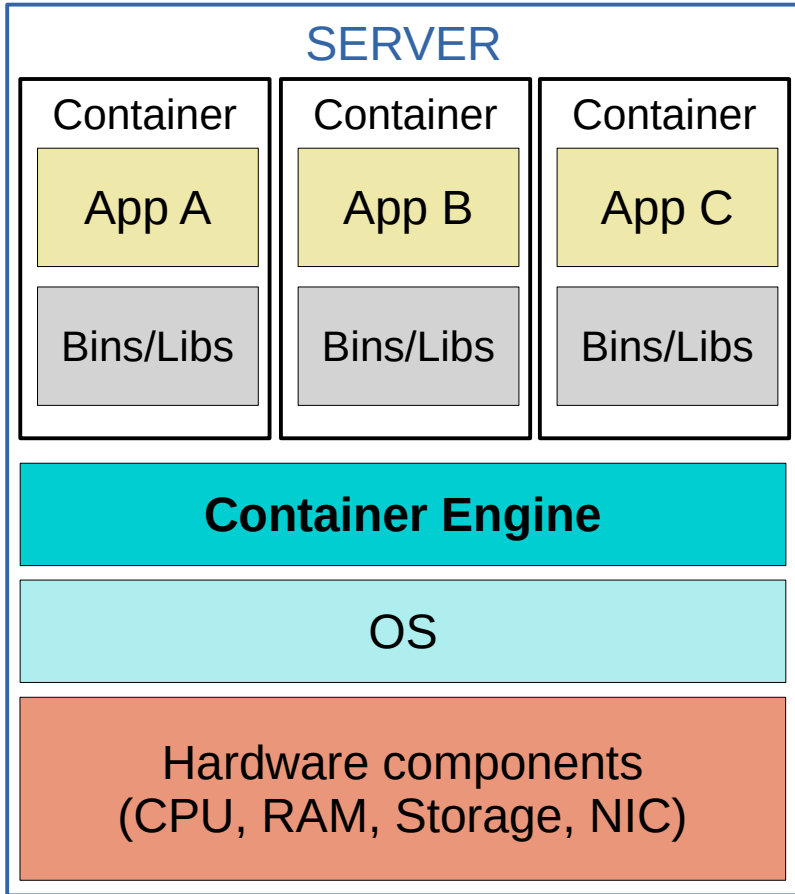
TYPE 1

native
(bare metal)



TYPE 2

hosted



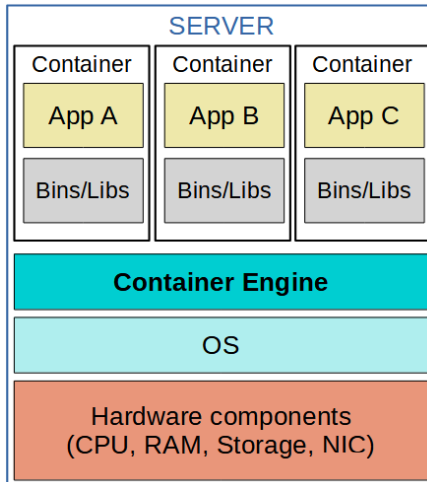
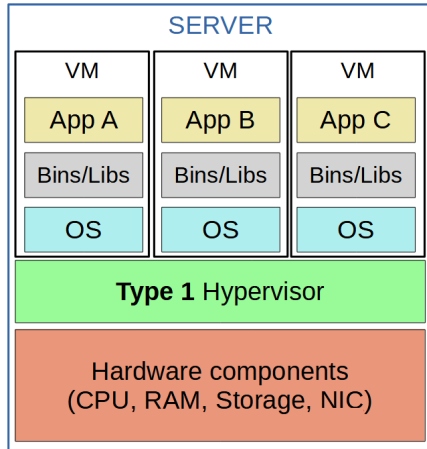
- **Containers** are software packages that contain an App and all dependencies (*Bins/Libs* in the diagram) for the contained App to run.
→ Multiple Apps can be run in a single container, but this is not how containers are usually used.



- Containers run on a **Container Engine** (ie. Docker Engine)
→ The container engine is run on a host OS (usually Linux).
- Containers are lightweight (small in size) and include only the dependencies required to run the specific App.
- A **Container Orchestrator** is a software platform for automating the deployment, management, scaling etc. of containers.
→ **Kubernetes** (originally designed by Google) is the most popular container orchestrator.
→ **Docker Swarm** is Docker's container orchestration tool.
- In small numbers manual operation is possible, but large-scale systems (ie. with *Microservices*) can require thousands of containers.

Microservice Architecture is an approach to software architecture that divides a larger solution into smaller parts (microservices).
→ Those microservices all run in containers that can be orchestrated by Kubernetes (or another platform).

VMs vs Containers



- **VMs** can take minutes to boot up as each VM runs its own OS.
- **Containers** can boot up in milliseconds.
- **VMs** take up more disk space (gigabytes).
- **Containers** take up very little disk space (megabytes).
- **VMs** use more CPU/RAM resources (each VM must run its own OS).
- **Containers** use much fewer CPU/RAM resources (shared OS).
- **VMs** are portable and can move between physical systems running the same hypervisor.
- **Containers** are more portable; they are smaller, faster to boot up, and Docker containers can be run on nearly any container service.
- **VMs** are more isolated because each VM runs its own OS.
- **Containers** are less isolated because they all run on the same OS; if the OS crashes, all containers running on it are effected.

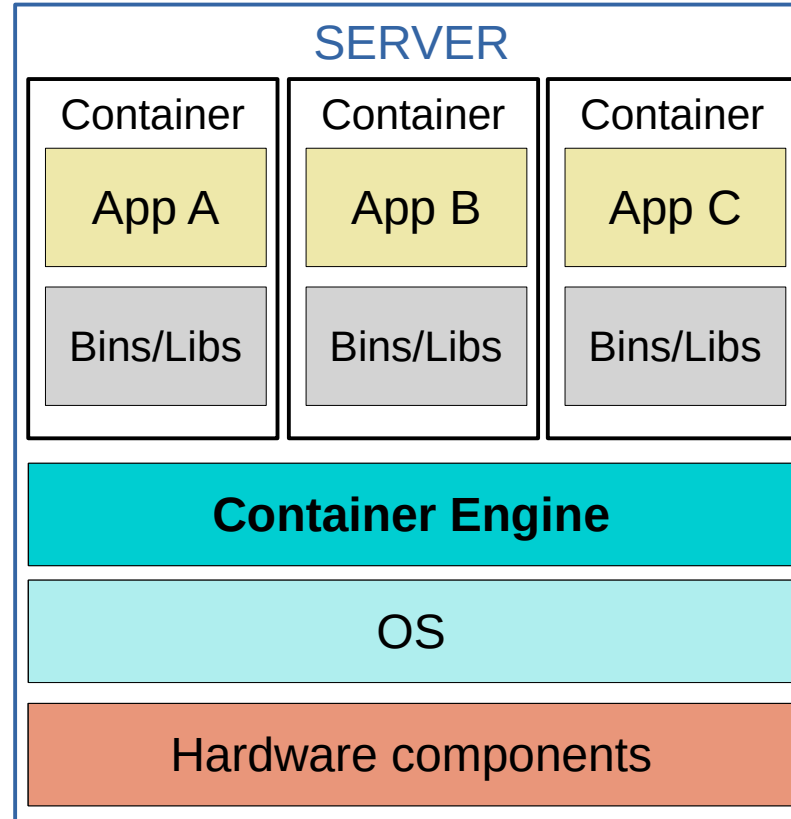
There is a major movement toward the use of containers, especially with the rise of microservices, automation, and DevOps (the combination of Software **Development** and IT **Operations**), but VMs are still widely used today.

Things we covered

- Review of Virtual Machines
 - Type 1/Type 2 Hypervisors
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- Virtual Machines vs. Containers

Quiz 1

Identify the three components below that containers run on top of.



Which of the following are examples of container orchestrators? (select two)

- a) Docker Engine
- b) Docker Swarm
- c) Kubernetes
- d) Hyper-V

Which of the following statements about VMs and containers are true? (select three)

- a) VMs require more resources.
- b) Containers take more time to boot up.
- c) VMs are more isolated.
- d) An OS runs in each container.
- e) Containers are often tens of gigabytes in size.
- f) Containers all run on a host OS.